

PIXEL SENSE

AI-powered smart eyewear that's dedicated to
make the life of visually impaired people
interactive and easier.

LISTEN TO THE WORLD

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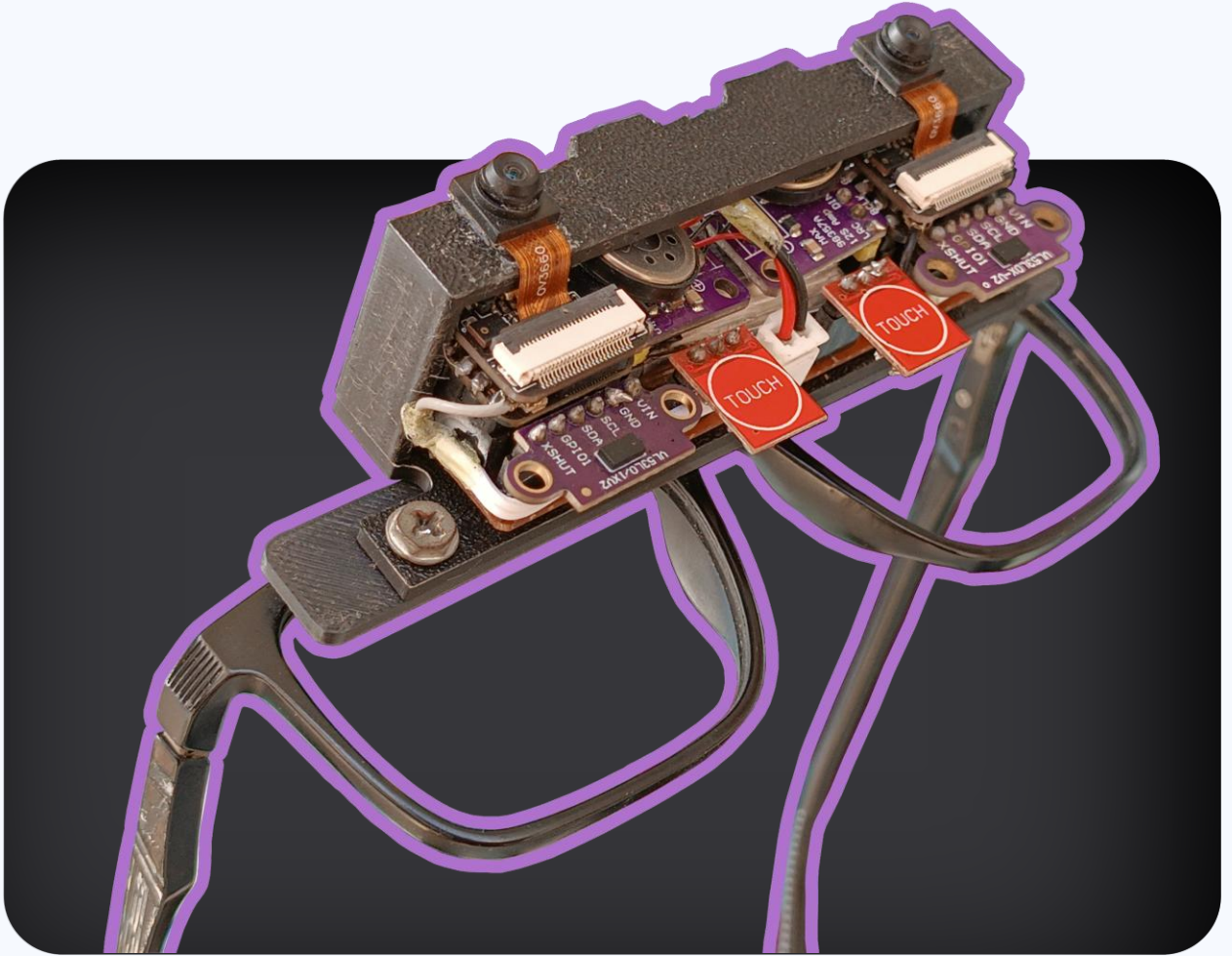
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Project Overview



PixelSense is a wearable assistive technology project designed to enhance the lives of **visually impaired individuals**. Integrated into eyeglasses, it features two cameras for stereo vision, laser distance sensors, microphones, stereo speakers, internet-connected MCU, and user-interaction buttons.

This project aims to enhance the independence of visually impaired individuals by enabling them to perceive their environment through spatial audio and descriptive instructions.

Features

Depth Perception

Enables users to navigate their surroundings through real-time auditory environmental mapping. By leveraging **dual-lens stereoscopic cameras**, the device continuously generates a high-precision **absolute depth map**. This visual data is then translated into a multi-dimensional **spatial audio landscape** delivered via integrated stereo speakers. The spatialization algorithm encodes the three-dimensional environment using the following parameters:

1. **X-Axis (Horizontal Localization):** The horizontal field of view is mapped through **interaural level differences (ILD)**. The system performs a rhythmic "sweep" across the scene, transitioning the audio intensity from the left to the right channel to provide a clear sense of lateral positioning.
2. **Y-Axis (Vertical Elevation):** Verticality is represented through a **frequency-mapped pentatonic scale** consisting of 20 distinct bands. The pitch modulates from a fundamental frequency at the top of the frame to a perfect fifth, reaching a full octave at the bottom, allowing the user to distinguish height through tonal shifts.
3. **Z-Axis (Proximity & Distance):** Depth is communicated through **dynamic volume modulation** governed by an **inverse-exponential function**. This ensures that distant objects produce a subtle auditory footprint, while immediate obstacles trigger a significantly louder signal, providing a natural and intuitive sense of urgency for nearby hazards.



Freeform

Enables users to understand their environment through conversational descriptions. The device captures an image, processes user-posed questions (via microphone) about the surroundings, and delivers answers (via inbuilt speakers) using an LLM.



Coordination

Provides object location relative to the user's hand in real-time. It utilizes a **Vision Transformer (ViT) tracker** to provide real-time spatial guidance. By performing a continuous **geometric analysis of bounding boxes** for both the hand and the designated object (The user utters the name of the object to the inbuilt microphone), the system calculates precise directional vectors. These coordinates are then translated into **frequency-coded auditory feedback**, providing the user with low-latency acoustic cues to facilitate intuitive manual interaction and successful object retrieval.



Text Recognition

Reads text aloud. The device captures an image of text (books, signs, handwriting), performs **Optical Character Recognition (OCR)**, and vocalizes the content through the inbuilt speakers.



Proximity Alert System

Provides high-speed obstacle detection by utilizing two integrated **Time-of-Flight (ToF)** sensors to facilitate real-time obstacle avoidance. Operating with a **50ms latency**, the module ensures rapid response times for identifying immediate physical barriers, such as walls, to prevent collisions. The system employs **frequency-modulated auditory alerts**, where the acoustic pitch dynamically shifts in relation to proximity.

Problem Addressing

Visually impaired individuals face significant daily challenges related to spatial navigation, environmental awareness, object interaction, and access to written information. Traditional assistive tools, such as white canes or guide dogs, provide vital mobility support but offer limited contextual or three-dimensional feedback.

PixelSense is engineered to bridge these critical sensory gaps. By translating complex visual data into intuitive auditory cues and conversational feedback, the system addresses the following specific challenges:

Lack of 3D Spatial Awareness and Safe Navigation

The Challenge: Visually impaired people often struggle to perceive the broader layout of a room, the height of overhanging objects, or the distance of approaching hazards, leading to a restricted sense of space and potential collisions.

The Solution: PixelSense provides a continuous, high-precision understanding of the environment through its **Depth Perception** and **Proximity Alert** systems.

- The stereoscopic cameras map the environment across X, Y, and Z axes, translating lateral position, vertical elevation, and proximity into a rich, multi-dimensional spatial audio landscape.
- For immediate, fast-approaching hazards (e.g., a closing door or an unseen wall), the dual Time-of-Flight (ToF) sensors provide a high-speed, 50ms-latency frequency-modulated auditory alert, preventing bodily collisions before they happen.

Difficulty Grasping and Locating Specific Objects

The Challenge: Finding a specific item—such as a mug on a cluttered table, a dropped set of keys, or a specific tool—often requires "tactile fumbling." This method of feeling around is not only frustrating and time-consuming but can also be dangerous if the object is sharp, hot, or fragile.

The Solution: The **Coordination** feature eliminates the need for tactile searching. By uttering the name of an object, the Vision Transformer (ViT) tracker tracks the user's hand in real-time. Through dynamic, frequency-coded acoustic vectors, PixelSense guides the user's hand precisely to the target, restoring seamless manual dexterity and object retrieval.

Disconnect from Complex Environmental Context

The Challenge: Safely navigating a space does not necessarily mean understanding it. Visually impaired individuals often miss out on the contextual nuances of their surroundings (e.g., "Is this store crowded?", "What kind of room am I standing in?", or "What is on the desk in front of me?"). Relying on sighted individuals for this context limits autonomy.

The Solution: The **Freeform** feature acts as an on-demand, conversational visual assistant. By combining camera capture with a Large Language Model (LLM), users can ask complex, open-ended questions about their surroundings and receive descriptive, spoken answers, granting them profound contextual independence.

Inaccessibility of Printed Text in the Physical World

The Challenge: We live in a text-heavy world. Menus, street signs, mail, pill bottles, and books present constant barriers to independent living, forcing visually impaired individuals to rely heavily on sighted assistance, braille alternatives, or cumbersome smartphone scanning apps.

The Solution: The **Text Recognition** module completely automates the reading process. By utilizing high-accuracy Optical Character Recognition (OCR), PixelSense seamlessly captures both printed and handwritten text in the user's field of view and reads it aloud, granting autonomy.

Hardware Details

Component	Purpose
2x Esp32 S3 MCU	Connect to server, Audio processing and streaming, image processing and streaming, reading TOF and touch sensors and connecting everything together.
2x OV3660 3MP camera	Capturing images
2x MSM261D3526HICPM digital MEMS microphone	Capturing audio samples and sending to ESP32 via I2S for user interactions
2x MAX98357A DAC + Amplifier	Playing audio samples received from ESP32 via I2S
2x VI53L0X TOF infrared-laser distance sensor	Low-latency obstacle avoidance
2x TTP223 touch sensors	User interactions
500mAH Battery	Powering the whole system

Fig: Schematic Diagram

Server-side (Back-end) Details

The assistive device can't process all the heavy AI tasks like object detection, stereo depth mapping, OCR, speech to text and text to speech etc. locally as the hardware capabilities are limited. The server, connected to the internet is used to perform heavy duty tasks, while also keeping the actual device cheap for accessibility.

Our server is based on python and connects to the individual ESP32 via WebSocket. Here, to transmit data between the ESP32 and server, we have utilized two network protocols:

TCP: Establishing connection through handshake. Sending and receiving small commands and information like TOF distance reading, featuring execution order etc.

UDP: Used to stream audio and JPEG images bidirectionally between server and the ESP32.

Various core features are executed in the server:

Speech Recognition

Upon user's request of a features that requires speech recognition, the ESP32 streams audio samples at 8Khz, 16bit quality to the server via UDP. The server transcribes the audio in real-time through the **Tiny Whisper model** developed by OpenAI. The transcribed audio then moves to the next step for the specific feature.

The LLM

Features like Text recognition and Freeform heavily depend on LLM for providing human-like description of environment to the user. For superior text recognition and natural and effective text output, we have utilized the Gemini 3 Flash model. Upon receiving text from the LLM, we transcribe it into 12KHz, 16bit audio through **Piper text to speech** and stream the audio samples from server to ESP32. The ESP32 then plays the audio samples via the DAC and Amplifier modules.

Coordination Feature

The Coordination feature allows users to locate and grasp specific, everyday objects by uttering their names.

The Technical Challenge:

Generic object detection models (like YOLO) are limited to pre-trained categories and cannot recognize the vast, unpredictable array of household items. Conversely, while Large Language Models (LLMs) possess the "world knowledge" required to identify any described object, using them for continuous, live video tracking introduces latency that is too slow for real-time interaction.

The Solution:

PixelSense utilizes a hybrid architecture that balances the high accuracy of an LLM with the low latency of a Vision Transformer (ViT) tracker and MediaPipe.

The process operates in three streamlined phases:

Initialization (LLM): When the user names an object, a single camera frame is sent to an LLM (Gemini 2.5 Flash, optimized for spatial awareness). The LLM locates the requested item and generates precise initial bounding box coordinates (x, y, width, height).

Real-Time Tracking (ViT): This initial bounding box is fed to a high-speed Vision Transformer (ViT) tracker. Using the LLM's bounding box as a visual reference, the ViT takes over to continuously track the object in real-time without the computational overhead of an LLM.

Spatial Guidance: Simultaneously, MediaPipe algorithms continuously track the user's hand. The system calculates the directional vector between the hand and the ViT-tracked object. These coordinates are sent to the ESP32, which generates dynamic, frequency-coded audio cues to guide the user's hand directly to the target.

Depth Map Estimation

PixelSense utilizes dual cameras to generate a high-precision understanding of the user's environment. To balance power consumption on the wearable device, the heavy computation is offloaded to a server using a streamlined computer vision pipeline:

Synchronous Data Streaming: Upon activation, the two ESP32 microcontrollers capture and stream synchronized JPEG image pairs (at 640x480 resolution) to the server via UDP.

Stereo Rectification: The server receives the synchronized frames and aligns them using OpenCV's stereo rectification algorithm. This step utilizes pre-calculated matrices derived from standard chessboard pattern calibration.

Disparity Calculation: The rectified image pairs are fed into a Stereo Semi-Global Block Matching (SGBM) algorithm. This calculates the pixel disparity (the optical shift of objects) between the left and right camera views.

Map Refinement: The raw disparity map is then processed through a Weighted Least Squares (WLS) filter to reduce noise, smooth edges, and improve overall accuracy.

Final Depth Generation: Finally, the system calculates the absolute 3D depth map by combining the refined disparity map with the hardware's pre-calibrated physical parameters (the baseline distance between the two lenses and their focal lengths).

